

Claim Support Log

Use this file to track dissertation claims, the literature that supports them, and whether more literature is needed.

Claim 1 - Trading fours provides a structured model for musical dialogue

Claim

Trading fours is useful for Intelli-Trading Fours because it structures improvisation as alternating short turns in which listening, interpretation, and response are mutually dependent.

Status

Well supported.

Best Supporting Sources

Source	How It Supports The Claim	Strength
Donnay et al. 2014	Directly studies trading fours as interactive musical improvisation and frames it as collaborative musical communication.	strong
Hodson 2007	Discusses jazz improvisation as interaction, interplay, and musical conversation.	strong
Hoffman and Weinberg 2011	Supports call-and-response as an evaluable musical interaction task.	moderate

Needs More Literature?

No, unless a deeper jazz-specific trading-fours source is desired.

Dissertation Location

Chapter 1.1; Chapter 2.5.

Notes

Use Donnay et al. for the direct trading-fours claim and Hodson for the broader jazz conversation/interplay framing.

Claim 2 - Turn-based interaction reduces ambiguity compared with continuous co-improvisation

Claim

Segmenting interaction into explicit turns makes the system easier to structure computationally because capture, analysis, planning, playback, and re-entry can be represented as distinct phases.

Status

Partially supported.

Best Supporting Sources

Source	How It Supports The Claim	Strength
Donnay et al. 2014	Supports trading fours as structured alternation in interactive improvisation.	strong for musical basis
Petit and Serrano - Skini	Supports state-based/reactive structuring of interactive music.	moderate for CS architecture
Herrington 2010	Supports layered sensing-processing-response design in interactive music systems.	moderate
Rowe 1991	Likely relevant for real-time agents and machine listening, but needs OCR/manual check.	pending

Needs More Literature?

Maybe. One additional HCI or interactive music source specifically about turn-taking, dialogue systems, or interaction decomposition would strengthen this.

Dissertation Location

Chapter 1.1; Chapter 3.2.

Notes

Avoid claiming that the literature proves turn-taking is always superior. Phrase it as a practical design choice for this system.

Claim 3 - IT4s should be evaluated as an interactive process, not only as generated output

Claim

The success of IT4s depends on the responsiveness, timing, interpretability, and perceived musical exchange of the system, not only the standalone quality of generated rhythms.

Status

Well supported.

Best Supporting Sources

Source	How It Supports The Claim	Strength
Smith 2022	Frames AI music systems through partnership, agency, and performer experience.	strong
Ostermann et al.	Evaluates a reactive jazz accompaniment system through user study and creativity/perception criteria.	strong
Hoffman and Weinberg 2011	Evaluates embodiment and response accuracy in an interactive musical task.	strong
Pasquier et al. 2016	Places interactive systems within musical metacreation.	moderate

Needs More Literature?

No for dissertation-level framing. More would only be needed for a full user-study methodology.

Dissertation Location

Chapter 2.7; Chapter 7; Chapter 8.6.

Notes

Use this claim to justify tests, diagnostics, debug tooling, and future drummer evaluation.

Claim 4 - Rhythm can be represented as a structured grid or cyclic pattern

Claim

Quantised rhythmic representations are defensible because rhythm can be modelled computationally as a structured sequence of onsets and silences over a temporal cycle or grid.

Status

Well supported.

Best Supporting Sources

Source	How It Supports The Claim	Strength
Toussaint 2004	Directly represents rhythm as cyclic binary strings.	strong
Longuet-Higgins and Lee 1984	Discusses rhythmic interpretation and metrical ambiguity.	strong
Gillick et al. 2019	Uses quantised drum representations and inverse transformations.	moderate

Needs More Literature?

No.

Dissertation Location

Chapter 2.4; Chapter 4.4; Chapter 4.5.

Notes

Make clear that IT4s adds velocity, sample timing, and turn bounds beyond a simple binary representation.

Claim 5 - Metrical salience is a defensible basis for anchor detection and skeleton generation

Claim

Rhythmic response planning can give special status to metrically or perceptually salient events because rhythmic interpretation depends on metrical structure, grouping, and temporal regularity.

Status

Well supported.

Best Supporting Sources

Source	How It Supports The Claim	Strength
Longuet-Higgins and Lee 1984	Discusses metrical interpretation, regularity, grouping, and ambiguity.	strong
Temperley 2009	Treats metrical analysis as a core symbolic music-analysis task.	strong
Levitin, Grahn, and London 2018	Reviews rhythm, meter, entrainment, pulse, and movement.	strong
Toussaint 2004	Supports structural analysis of onset placement in rhythmic cycles.	moderate

Needs More Literature?

No, although a rhythm salience-specific source could be added if the anchor chapter becomes very theory-heavy.

Dissertation Location

Chapter 2.6; Chapter 5.4; Chapter 6.4.

Notes

This supports the concept, not every weighting choice in [AnchorAnalyser](#). Specific weights should be framed as implementation choices.

Claim 6 - The current response-generation layer should be framed as interim

Claim

IT4s implements response planning and prototype structural generation, but the final full response-realisation layer remains future work.

Status

Well supported by comparison sources and codebase evidence.

Best Supporting Sources

Source	How It Supports The Claim	Strength
Pachet 2002	Shows a mature example of interactive style-aware continuation.	strong comparison
Martin and Torresen 2019	Shows learning-based predictive interaction as a more advanced alternative.	strong comparison
Hu et al.	Shows dataset/model-based call-response generation.	strong comparison
Gillick et al. 2019	Shows learned expressive drum generation/humanization.	strong comparison
Toussaint 2010	Shows algorithmic rhythm generation ideas for future constraints.	moderate comparison

Needs More Literature?

No.

Dissertation Location

Chapter 6.5; Chapter 6.6; Chapter 8.3; Chapter 8.5.

Notes

This claim should rely primarily on codebase evidence, with literature used as comparison and future-work context.
